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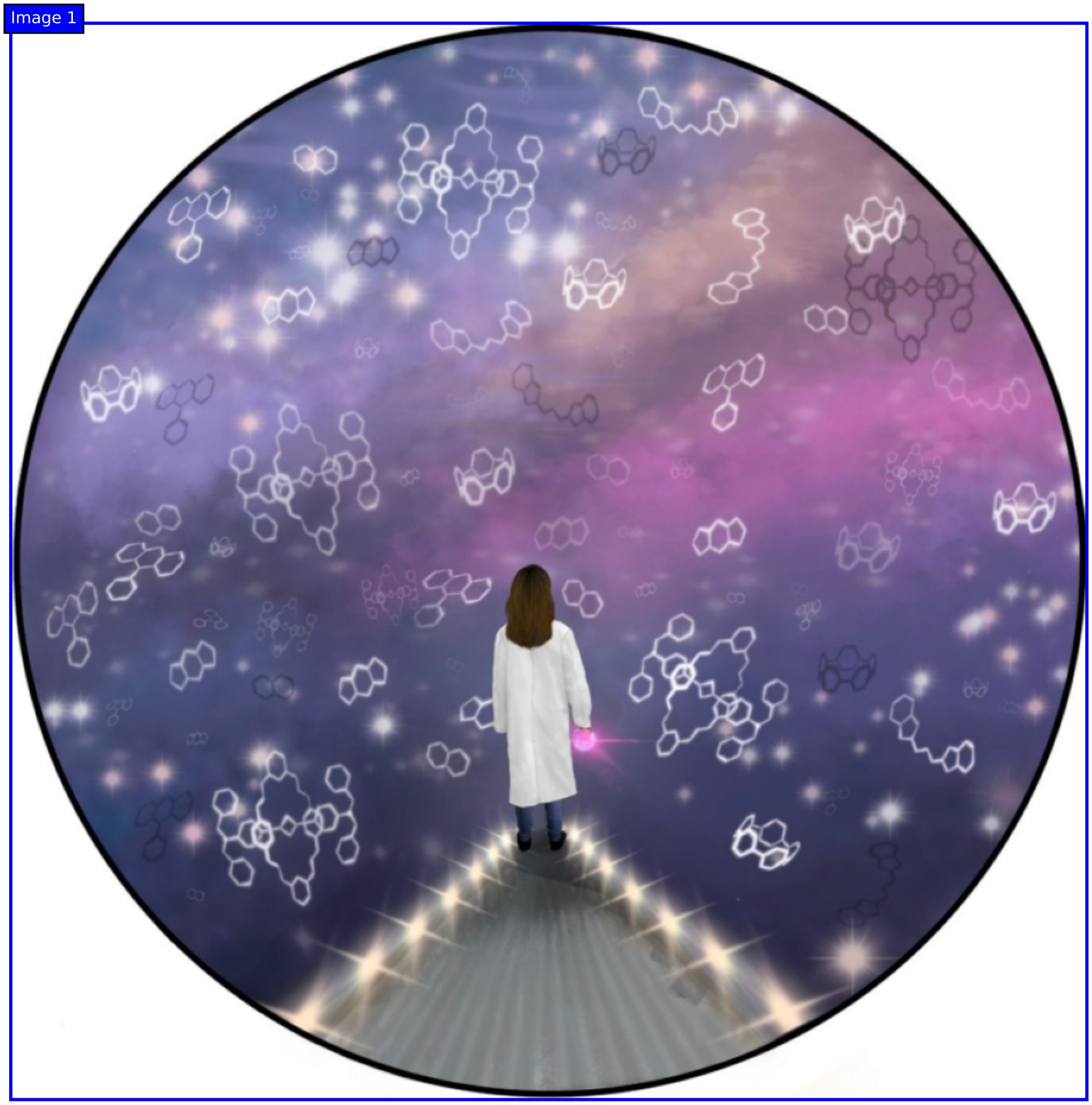
EurJOC
European Journal of Organic Chemistry

Review
doi.org/10.1002/ejoc.202301196

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An Organic Chemist's Guide to Fluorophores – Understanding Common and Newer Non-Planar Fluorescent Molecules for Biological Applications

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Text 1
Labeling and detection of biomolecules *in vitro* and *in vivo* is essential to many areas of biomedical science. Fluorophores stand as indispensable tools within chemical biology, underscoring the importance of fine-tuning their optical properties. This review focuses on methods for optimizing emission wavelength, quantum yield and photostability. We focus not just on the trends, but the fundamental physical organic chemistry

Text 2
concepts that inform the connection between molecular structure and fluorescent properties. This approach offers an essential understanding of fluorescence, enabling readers to develop a systematic analytical framework for thinking about fluorescence. Furthermore, an evaluation of newer non-planar fluorophores shines light on the bright future of fluorescent molecules.

Text 3
1. Introduction

Text 4
Albert Coons realized the utility of fluorescent molecules for biological applications when he labelled antibodies with fluorescein isothiocyanate in 1949.[1] Since then, fluorophores have been used in countless biological applications. Attachment to biomolecules allows interrogation of living systems, guidance for surgery, and enables disease identification. Moreover, fluorophores play a crucial role in numerous diagnostic kits, including those utilized in polymerase chain reaction (PCR) tests for diagnosing coronavirus (COVID-19). Interrogating disease progression requires simultaneous monitoring of multiple entities. Therefore, observing complex biological processes is intimately connected to the rational design and synthesis of bright, stable fluorescent molecules across the visible and near-infrared spectrum to provide multiple channels (colors) for multiplex biomarker analysis, allowing us to observe intricate biological processes.

Text 5
Currently, common commercially available synthetic fluorophores for biological applications rest on four foundational cores: coumarins, cyanines, xanthenes and boron dipyrromethenes (BODIPYs). While these scaffolds have undoubtedly provided useful fluorophores, some have been exploited for over a century. Therefore, we also include less utilized planar (perylene) and non-planar (cycloparaphenylene and squaraine-rotaxane) fluorophores. We delve into methods for optimizing the fluorescent attributes of these molecules, emphasizing not only discernible trends but also the essential tenets of physical organic chemistry that underlie their fluorescence. By exploring

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these concepts, we aim to provide a detailed tutorial for understanding fluorescence. Additionally, examining the unique photophysical characteristics of emerging fluorophores illuminates the promising future of fluorescent molecules in detection and sensing applications.

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1.1. The Fluorescence Process

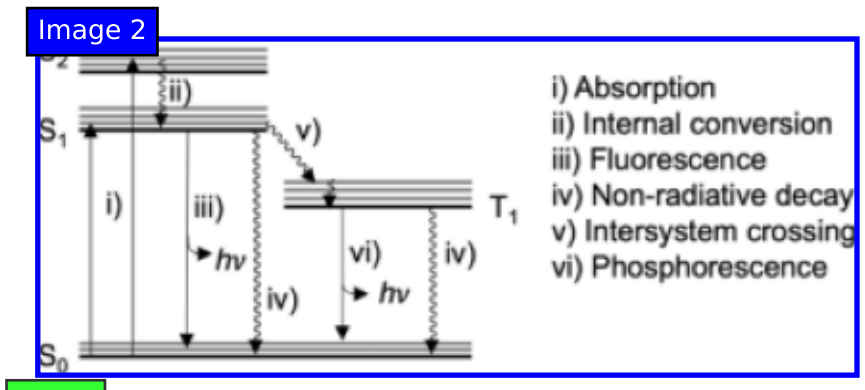
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The concept of fluorescence was first documented in 1845 by Sir John Herschel during his work with the antimalarial drug quinine. He observed a distinctive "celestial blue colour" emitted by quinine under certain lighting conditions.[2] It was not until 1852 when Sir George Stokes used quinine to work out the process of fluorescence.[3] This process is shown in the Jabłoński diagram in Figure 1.[4]

Text 9
Absorption of a photon promotes an electron from the ground state (S_0) to an excited state (S_1, S_2, S_3, etc.; Figure 1, i). Following absorption, energy is rapidly dissipated, leading to relaxation to the first singlet excited state (S_1; Figure 1, ii). Transition back to the ground state can occur through many paths. Fluorescence occurs when the molecule emits a photon ($S_1 \rightarrow S_0$; Figure 1, iii), but the molecule may return to the ground state through nonradiative pathways (Figure 1, iv), such as bond rotation or vibration.[5] Furthermore, in the excited state the molecule could undergo intersystem crossing to the triplet state (Figure 1, v).

Text 10
One key descriptor of the above processes is the extinction coefficient (ε, reported in $M^{-1}\,cm^{-1}$), which describes how well a molecule absorbs light and varies with the excitation wavelength. Common fluorophores have extinction coefficients ranging from 12,000 $M^{-1}\,cm^{-1}$ to 200,000 $M^{-1}\,cm^{-1}$.[6] Other important descriptors are the quantum yield (ϕ), representing the fluorescence efficiency, and fluorescence lifetime (τ), which is the average time interval between excitation and emission. The lifetime provides insight into both the intrinsic properties

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Image 2


Text 18
Figure 1. Jabłoński diagram illustrating electronic energy level transitions and non-radiative processes when exciting a fluorophore.

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Text 1

of the fluorophore (rates of fluorescence and competing non-radiative processes) as well as interactions between the fluorophore and its environment. The Stokes shift, denoting the energy difference between absorption and emission maxima within the same electronic transition, is crucial. A larger Stokes shift is favorable as it minimizes interference between incident and emitted light, reducing background noise and expanding the usable bandwidth for fluorescence collection. Brightness serves as a valuable comparator for fluorophores and is determined by the product of the extinction coefficient and quantum yield ($\varepsilon \times \phi$). High brightness is desirable for achieving a large signal-to-noise ratio and a lower detection limit. These fluorophore properties are a direct result of their molecular structure, which can be altered through structural manipulation to suit the needs of specific applications.

Text 2

For most biological applications, the ideal fluorophore would possess a quantum yield of 1, high brightness, emit in the red to near-infrared range, exhibit resistance to photobleaching even under high illumination power, maintain a small size to minimize disruption to biological systems, show no toxicity, and readily dissolve in aqueous environments. While achieving the truly ideal fluorophore remains elusive, efforts are directed toward the following objectives: i) shifting excitation and emission towards the red to mitigate cell damage, ii) enhancing extinction coefficients and quantum yields for improved brightness, and iii) reducing reactivity to reactive oxygen species (ROS) and shorten the triplet-state lifetime to enhance photostability.[7] This discussion outlines the strategies and approaches aimed at optimizing these objectives, ultimately advancing us closer to the quest for the ideal fluorophore.

Text 3

2. Tuning Fluorophore Emission

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To effectively monitor complex biological systems, it is necessary to observe multiple entities simultaneously. Therefore, fluorophores with varying emission wavelengths spanning the color spectrum are required. Furthermore, specific applications may require fluorophores with certain emission wavelengths. For instance, red-emitting fluorophores are advantageous in biological imaging because longer wavelengths experience less scattering and absorption in tissues allowing deeper penetration and imaging.[8] Additionally, longer wavelength (lower energy) light is also less damaging to cells. Therefore, controllably manipulating fluorophore optical properties is crucial. While there are numerous fluorophore reviews that list variants of common scaffolds and their fluorescent properties, they lack a thorough analysis of the underlying reasoning for these trends. Here, we provide a rational understanding based on fundamental principles of physical organic chemistry. Our goal is to assist researchers in tuning the optical properties of fluorescent dyes.

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Fluorophore emission is tuned by modifying the energy levels of frontier molecular orbitals (FMO), which include the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO). The smaller the HOMO-LUMO gap, the more red the emission wavelength. FMO theory is based on the linear combination of atomic orbitals (LCAOs),

Image 1



Text 7

Dr. Terri Lovell earned her BSc from Memorial University of Newfoundland and MSc in polymer science at the University of Oregon (UO). She worked at Thermo Fisher Scientific (legacy Molecular Probes) and became fascinated by fluorescent probes. She pursued a PhD with Prof. Jasti at UO; focusing on using nanohoops as new fluorescent scaffolds, fine-tuning their photophysical properties, and exploring their biological applications. Currently, she is an NSERC Postdoctoral Fellow at McGill University, working on improving fluorophores for single-molecule fluorescence microscopy.

Image 2



Text 8

Professor Ramesh Jasti earned his PhD at the University of California, Irvine, under Prof. Scott Rychnovsky. He discovered cycloparaphenylenes during a postdoctoral fellowship with Prof. Carolyn Bertozzi at the University of California, Berkeley. He expanded this work as an Asst. Prof. at Boston University (2009-2014). Currently, he is a Professor and Associate Director at the Materials Science Institute, University of Oregon. Prof. Jasti holds numerous patents and has received several honors including the NSF CAREER award, a Sloan Fellowship, and the Camille Dreyfus Teacher-Scholar award.

Image 3



Text 9

Bruce P. Branchaud received his Ph.D. in 1981 from Harvard University (R. B. Woodward mentor). He was an NIH-NRSA postdoctoral fellow 1981–1983 at the Massachusetts Institute of Technology (Christopher T. Walsh mentor). He is currently Distinguished Scientist in the Cancer Early Detection Advanced Research Center (CEDAR) in the Knight Cancer Institute at Oregon Health & Science University and Professor Emeritus in the Department of Chemistry & Biochemistry at the University of Oregon. His current work is focused on the use of organic chemistry / chemical biology to develop new and improved technologies for the early detection of cancer.

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where molecular orbitals are formed by combining atomic orbitals of individual atoms in a molecule.[9,10] This theory is used here to explain how electron withdrawing groups (EWG), electron donating groups (EDG), and extended π-conjugation affect the molecular orbitals and, therefore, emission. While it is important to acknowledge that the energy gap is influenced by multiple factors,[11,12] for simplicity, we focus on the HOMO and LUMO orbital densities at each position of the scaffold. This description is accurate for most dyes as they obey Kasha's rule, where emission is from the lowest lying excited state. However, there are a few exceptions to these assumptions, which are mentioned herein. This simplification does not accurately describe absorbance since absorbance can happen via many transitions, not just the HOMO to LUMO transition. We note that EDG and EWG affect FMO distribution, but we show that orbital distribution in the parent molecule predicts substituent effects well. Finally, it's worth noting that there exist methods beyond structural modifications that can induce a red-shift in fluorescence, such as J-aggregation and other solvatochromic techniques.[13–18] However, our primary focus lies in examining fluorophore structures and how the incorporation of substituents impacts the properties of these fluorophores.

2.1. Electron Withdrawing and Donating Groups Alter Molecular Orbitals

To induce a red-shift (bathochromic shift) in the emission of fluorophores, common methods involve the addition of EDGs or EWGs. In the simple case of benzene (Figure 2), addition of electron donating groups destabilize (raise) the HOMO and LUMO energies, whereas electron withdrawing groups have a stabilizing effect (lowering orbital energies).[11,19] By adding electron donor or acceptor groups to positions where only one FMO is localized, the energy level of one FMO can be electronically modulated without affecting other. This changes the HOMO-LUMO gap and, therefore, fluorescence.[12] Hammett parameters are primarily intended to quantify substituent electronic effects on the reactivity of organic molecules in chemical reactions.[20] Nevertheless, they are also frequently employed to assess a substituent's impact on the photophysical properties of a fluorophore. However, Hammett parameters do not provide a holistic picture of substituent effects on fluorescence, which is unsurprising since the parameters are derived from the ionization of organic acids in solution.[21]

Therefore, qualitative molecular orbital theory is used to understand the photophysical properties of conjugated molecules. Figures 3–7 display the atomic contributions to the HOMO and LUMO as percentages for each scaffold. Orbital contributions are only shown for positions amenable to functionalization and those displaying contributions exceeding 4%. Computational details and a comprehensive list of contributions can be found in the SI. Additionally, the figure captions list the corresponding solvents in which measurements were taken and these solvents are taken into consideration when analyzing trends, as the solvent can significantly affect the optical properties of the fluorophores.

Coumarin. Coumarin fluorophores, shown in Figure 3, have a phenol (**1**) or amine (**2**) on the 7-position. There is similar HOMO and LUMO orbital density on this position, but the HOMO electron density extends to the heteroatom. Therefore, switching the phenol to an electron donating amine results in a red-shift. There is also significant LUMO orbital density on the 4-position with minimal HOMO density. Therefore, adding EWG on the 4-position should stabilize (decrease) the LUMO, thereby narrowing the energy gap and causing a red-shift in fluorescence.[12] Substituting the methyl group in **2** for a CF_3 (**3**) results in a shift from 435 nm for **2** to 480 nm.[23] The CF_3 has a significant effect on the LUMO of **3**, but minimal impact on the HOMO, resulting in a further red-shifted emission compared to **2**. Replacing the methyl group in **2** with a carboxylic acid (**4**) in the 4-position causes a 60 nm red-shift.[24] **4** is further red-shifted than **3** due to conjugation of the carboxylic acid with the scaffold (see extending conjugation section). Introducing a

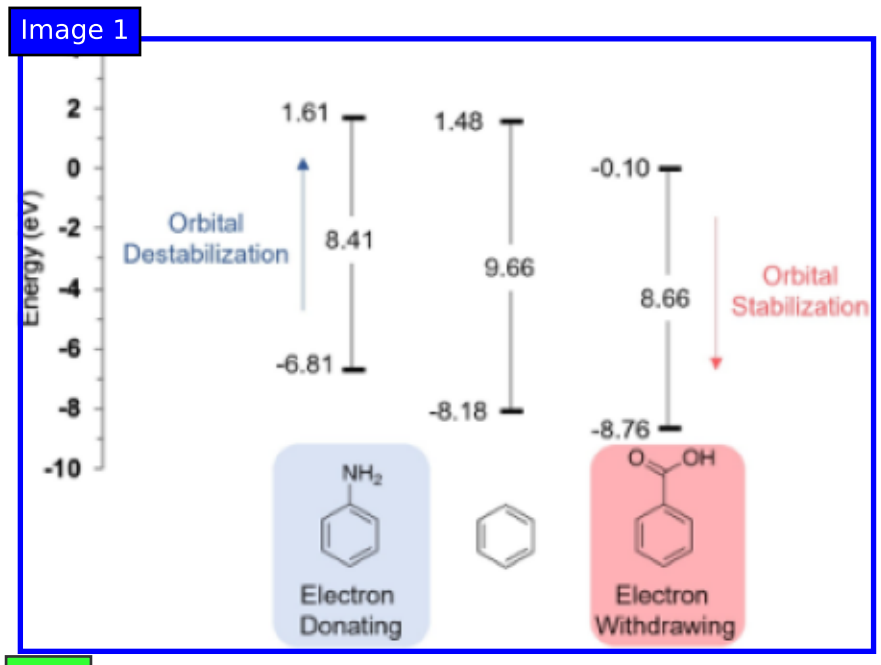


Figure 2. Orbital stabilization and destabilization from addition of electron withdrawing and electron donating groups. Orbital levels calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum.[22]

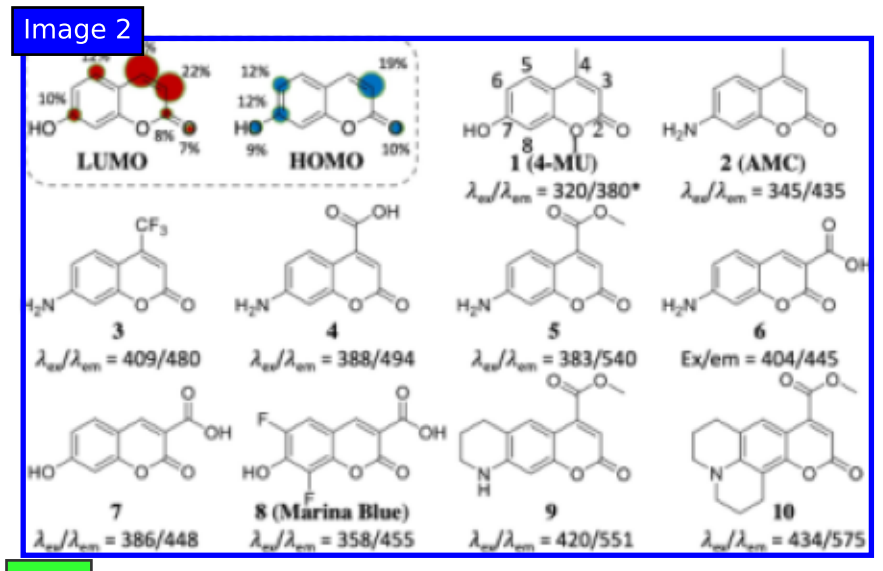


Figure 3. Coumarin HOMO and LUMO orbital densities and fluorescence tuning by incorporating EWGs and EDGs.[24] Orbitals calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum and contributions were visualized using Multiwfn.[26] **1**[27] is in methanol; **7** and **8**[25] at pH 10; **2**,[24] **3**,[23] **4**–**6**,[24] and **9**–**10**[24] in ethanol. *4-MU fluorescence listed is in methanol for the sake of comparison to AMC. The optimal solvent for 4-MU is aqueous base and the fluorescence is shifted to 450 nm.[25]

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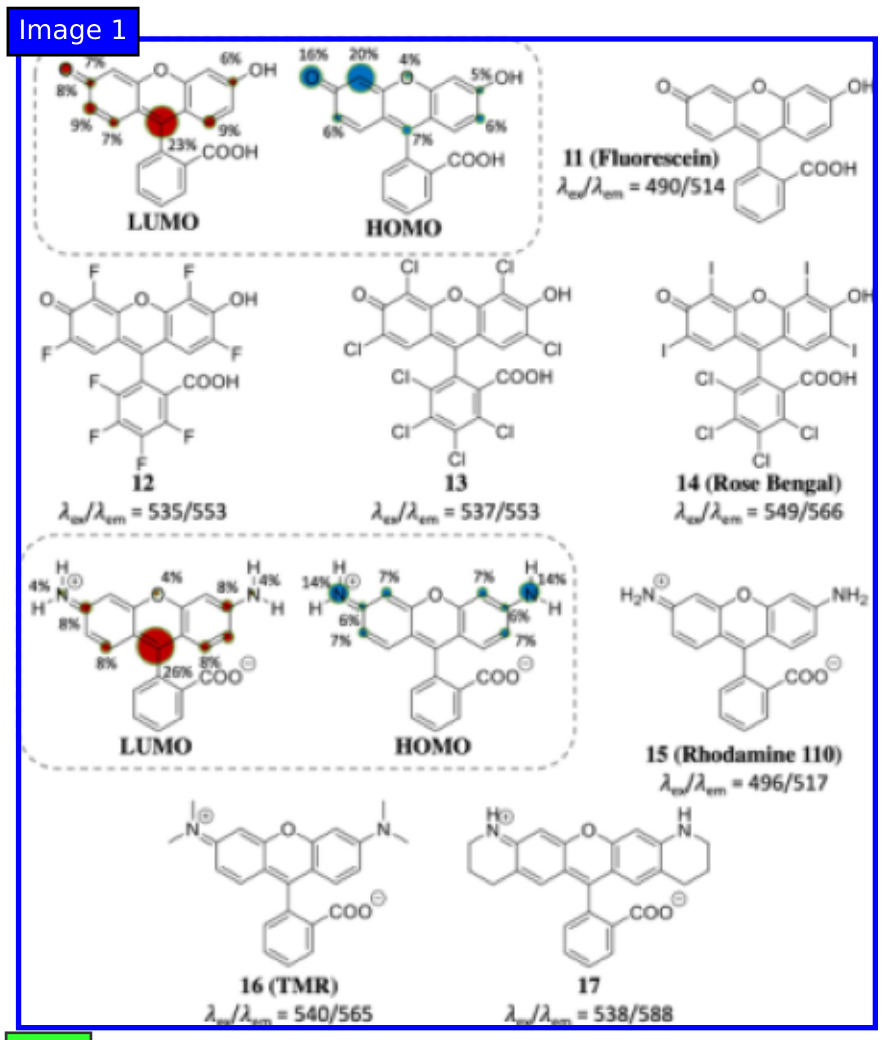
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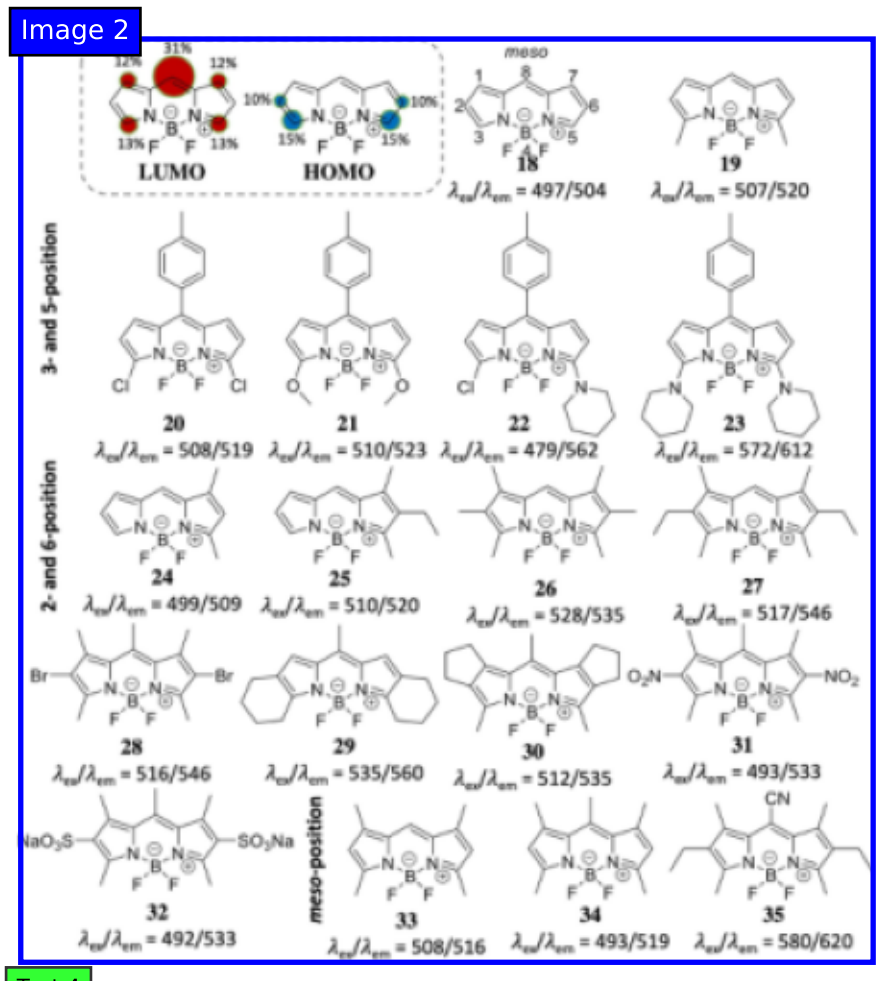
Image 1



Text 3

Figure 4. HOMO and LUMO orbital densities of fluorescein and rhodamine and fluorescence tuning by incorporating EDGs. Orbitals calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum and contributions were visualized using Multiwfn.[26] **11**,[4] **12**,[28] **13**,[29] **14**[29] at pH 9; **15**[30] at pH 7.5; **16**[31] in methanol; **17**[32] pH 7.3.

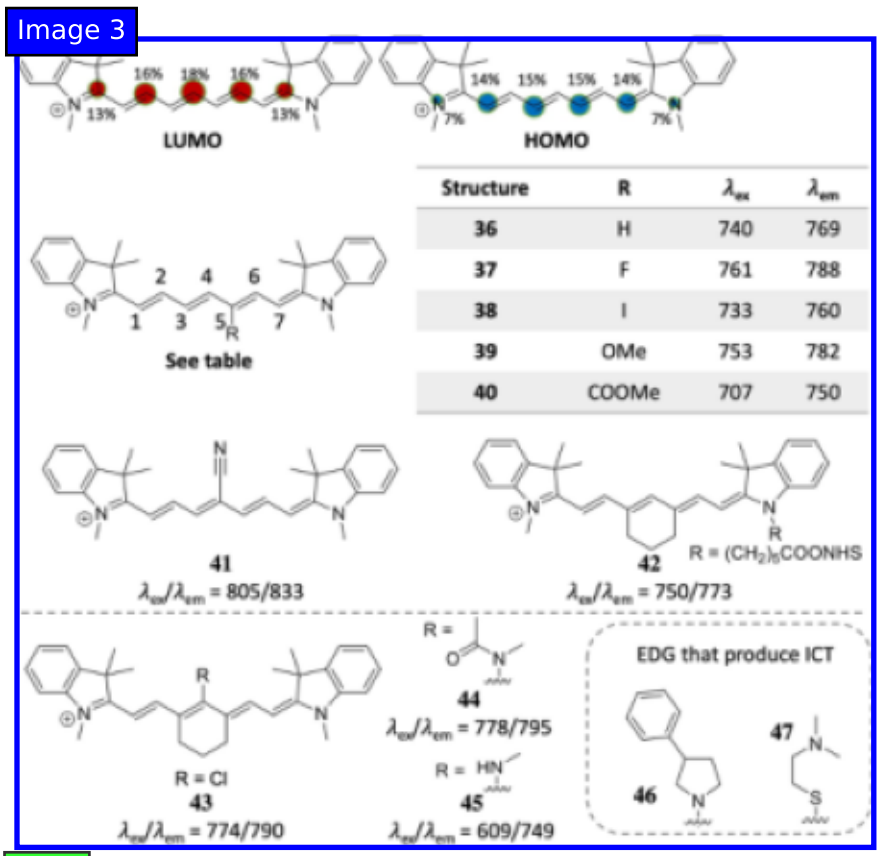
Image 2



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Figure 5. HOMO and LUMO orbital densities of BODIPY and fluorescence tuning by incorporating EWGs and EDGs. Orbitals calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum and contributions were visualized using Multiwfn.[26] **18**,[34] **20–23**,[35] **31–32**,[36] **34**[36] in methanol; **19**,[37] **24–27**,[37] **29–30**,[38] **33**,[37] **35**[38] in ethanol; **28**[36] in 9:1 methanol:dichloromethane.

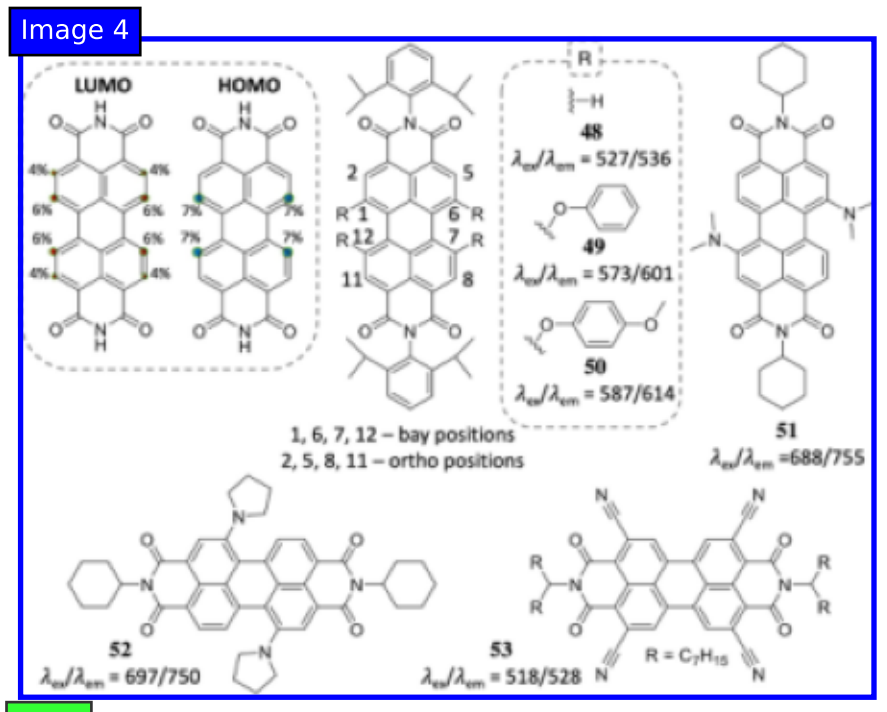
Image 3



Text 5

Figure 6. HOMO and LUMO orbital densities of Cyanines and fluorescence tuning by incorporating EWGs and EDGs. Orbitals calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum and contributions were visualized using Multiwfn.[26] **36–41**,[40] **42**,[41] **43–45**[42] in methanol; **46**,[43] **47**[44] in ethanol.

Image 4



Text 6

Figure 7. HOMO and LUMO orbital densities of PDI and fluorescence tuning by incorporating EWGs and EDGs. Orbitals calculated at CAM-B3LYP/6-31 + G(d,p) in a vacuum and contributions were visualized using Multiwfn.[26] **48**,[44] **49–50**[51] in toluene, **51–52**,[52] **53**[54] in dichloromethane.

Text 7

Text 1

methyl ester in the 4-position, as in **5**, induces a further red-shift in fluorescence compared to the carboxylic acid derivative **4** as the ester is more withdrawing. On the other hand, substituting the 3-position (**6**) with a carboxylic acid does not influence the emission because the EWG affects the HOMO and LUMO equally, resulting in no change in the energy gap.

Text 2

The 6-position has HOMO contributions, but no LUMO contributions, therefore, donors narrow the HOMO-LUMO gap. Addition of weakly π-donating fluorine atoms on **7** slightly red-shifts the fluorescence (**8**).[24,25] Halogens have competing

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Text 1
inductive and resonance, however, on phenylenes they act as weak EDGs.

Text 2
In amino coumarins, enhancing the nitrogen donation ability also leads to a red-shift in fluorescence. Formation of heterocycle **9** improves hyperconjugation, shifting the fluorescence to 551 nm. Fused systems further shift the fluorescence to 575 nm (**10**).[24] Albeit, the red shift in emission comes at a cost, where the quantum yields of **9** and **10** are almost zero. In summary, red-shifting the coumarin scaffold is achieved by putting EDGs on the 6- and 7-positions, and EWGs on the 4-position.

Text 3
Fluorescein and Rhodamine. Fluorescein and rhodamine HOMO and LUMO orbital densities are shown in Figure 4. The photophysical properties of fluorescein are more difficult to tune than rhodamine because substitution of the phenol in fluorescein yields a non-fluorescent lactone form.[6] Therefore, the main way to modulate fluorescein fluorescence is through halogenation.[6] Fluorination (**12**) and chlorination (**13**) of **11** prompts a modest bathochromic shift as they are weak π-donors.[4,28,29] Iodination (**14**) causes a further 10 nm shift.[29]

Text 4
However overall, the result is a mere 41 nm shift in fluorescence from **11**. The lack of fluorescein tunability shifted attention to the aminated rhodamine scaffold.

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Rhodamine 110, **15**, has an emission of 517 nm. When the donating ability of the amines are increased through methylation the fluorescence is shifted to 565 nm (**16**).[31] Heterocycle formation (**17**) provides optimal nitrogen lone pair conjugation with the π-system, red-shifting the fluorescence to 588 nm.[32] In summary, red-shifting xanthone dye fluorescence is achieved by halogenation of fluorescein or increasing the donating ability of rhodamine nitrogens. The most significant emission red-shift of the xanthene scaffold, however, comes from heteroatom incorporation (**Section 2.3**).

Text 6
BODIPY. Tuning BODIPY fluorescence has not been well understood in the past,[33] however, the fluorescence follows expected trends when using an FMO analysis. There are more HOMO orbital contributions on the 3- and 5-positions (Figure 5), indicating EDGs would result in a red-shift in fluorescence. Two EDGs in the 3- and 5-positions in **19** red-shifts the fluorescence from unsubstituted BODIPY **18**.[34] This effect is more significant when electron donating ability is increased. Switching chlorides of **20** to methoxy groups (**21**) only slightly red-shifts the fluorescence, but the effect is more dramatic with substitution of one (**22**) or two (**23**) amines.[35]

Text 7
BODIPY has HOMO density on the 2- and 6-positions and no significant LUMO contributions. Substituting an ethyl group on the 6-position as shown in **25** red-shifts the fluorescence compared to **24**.[37] The fluorescence is further shifted when substituting both the 2- and 6-positions with methyl (**26**) or ethyl groups (**27**). Substitution with bromide (**28**) shows a 42 nm bathochromic shift in fluorescence relative to **34**.[36]

Text 8
Furthermore, fusing cyclic structures on the 2- and 6-positions shifts the fluorescence further (**29**) due to better hyperconjugation.[38,39] However, if the cyclic structure is fused to positions with LUMO density, such as **30**, the same red-shifting as **29** is not observed.[38] There are a few structures that appear to stray from the expected trends. **31** and **32** have

Text 9
withdrawing groups on positions with mainly HOMO density, yet the fluorescence is red-shifted versus the unfunctionalized version **34**.[36] However, when considering the geometry of the molecule, the nitro group of **31** is parallel with the π-system and therefore is in conjugation. The resultant π-system extension results in the observed red-shift (see extending conjugation section). This is also true for sulfonate **32**.[36]

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Burgess stated the more the BODIPY core is functionalized, the further red it will emit.[33] However, if EDG are added to positions where the LUMO mainly resides, then the fluorescence is blue-shifted (hypsochromic shift). This is illustrated with **19** versus **33** and **34**.[36,37] **33** is more highly functionalized than **19**, but there is a weak EDG in the *meso*-position, which destabilizes the LUMO resulting in a slight blue-shift relative to **19**. **34** is even more functionalized than **19**, but the emission is the same. This reinforces the importance of considering orbital contribution on each position. The *meso*-position shows exclusively LUMO contribution. Therefore, adding EWG to the *meso*-position, as in **35**, red-shifts the fluorescence by over 60 nm compared to **27**.[38] In summary, red-shifting the BODIPY scaffold is achieved by putting strong EDGs on the 2-, 3-, 5-, and 6-positions and strong EWGs on the *meso*-position.

Text 11
Cyanine. Cyanines are composed of polycyclic aromatic moieties, typically indoles, linked by a polymethine bridge. They are commonly named as 'Cy' followed by a number that indicates the number of carbon atoms in the bridge, such as Cy3 and Cy5. Cyanines have alternating HOMO and LUMO density on the polymethine bridge. The LUMO density is on the even number positions and the HOMO is on the odd positions (Cy7 shown in Figure 6). Addition of electron donors such as fluorine (**36** to **37**) and methoxy (**39**) to an odd position destabilizes the HOMO, resulting in a fluorescence red-shift.[40]

Text 12
Unlike when iodine and bromine are added to phenyl rings, when they are added to polymethines (**38**) they act as electron withdrawing moieties (inductive effect wins). This distinction arises from the difference in the electronic structure and conjugation of these two systems. Phenyl rings are aromatic, meaning they possess a high degree of conjugation and electron delocalization, whereas polymethines are cross-conjugated, leading to less efficient electron delocalization. The type of conjugation is important to pay attention to when designing fluorophores or interpreting fluorescent properties. Electron withdrawing moieties elicit a blue shift relative to Cy7 (**40**).[40] EWG on the even positions stabilizes the LUMO and red-shifts the fluorescence (**41**). The stronger the withdrawing moiety, the further red the emission. Fluorination of cyanine phenyl rings should show no change in emission, which is the case for the fluorinated thiadicarbocyanine synthesized by Armitage and coworkers.[45] However, perfluorinated Cy3 prepared by Funabiki and coworkers[46] shows a 16 nm blue-shift relative to the parent Cy3. This initially unexpected difference arises because the entire indole moiety must be considered as the R-group. Fluorination of both indole and benzothiazole leads to a slight decrease in their HOMO and LUMO levels. Nonetheless, for indole, this decrease in FMOs results in a slightly larger HOMO-LUMO gap, whereas the energy gap for benzothiazole remains constant (Figure S1). These findings

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Text 1
highlight the importance of exercising caution and treating the aromatic heterocycle as an R-group when utilizing FMOs to assess and predict cyanine fluorescence.

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Rigidified Cy7 has the same orbital distribution. Halogenation provides a moderate red-shift (**42** to **43**), whereas a more withdrawing N-acetyl substituent (**44**) elicits a 20 nm emission red-shift.[41,42,47] Conversely, addition of a donating amine (**45**) promotes a blue-shift greater than 20 nm.[42] It is important to note that when the *meso*-EDG becomes too strong (**46** and **47**), the molecule experiences intramolecular charge transfer (ICT). In this phenomenon, the molecule undergoes a transfer of electronic charge within its structure, leading to a loss of its cyanine character and disrupting the observed trends.[44,47,48] In summary, red-shifting the cyanine scaffold is achieved by putting EDG on the odd positions of the polymethine bridge and strong EWGs on the even positions.

Text 3
Perylene. The perylenediimide (PDI) scaffold has minimal orbital separation (Figure 7). The nitrogen atoms are located on a nodal plane, therefore, substitution of the imide does not significantly change the optical properties.[49] The bay region has slightly more HOMO density, making it the only viable region to add EDGs. When switching hydrogen (**48**[50]) for better donors like **49** and **50** the fluorescence shifts by almost 80 nm.[51] More efficient donors such as **51** and **52** cause shifts over 100 nm.[52] Although, one has to be mindful of the bay substituents. If the groups are too large it will result in a twisting of the PDI core, potentially leading to a hypsochromic shift.[49] There are other "push-pull" variants of perylene lacking

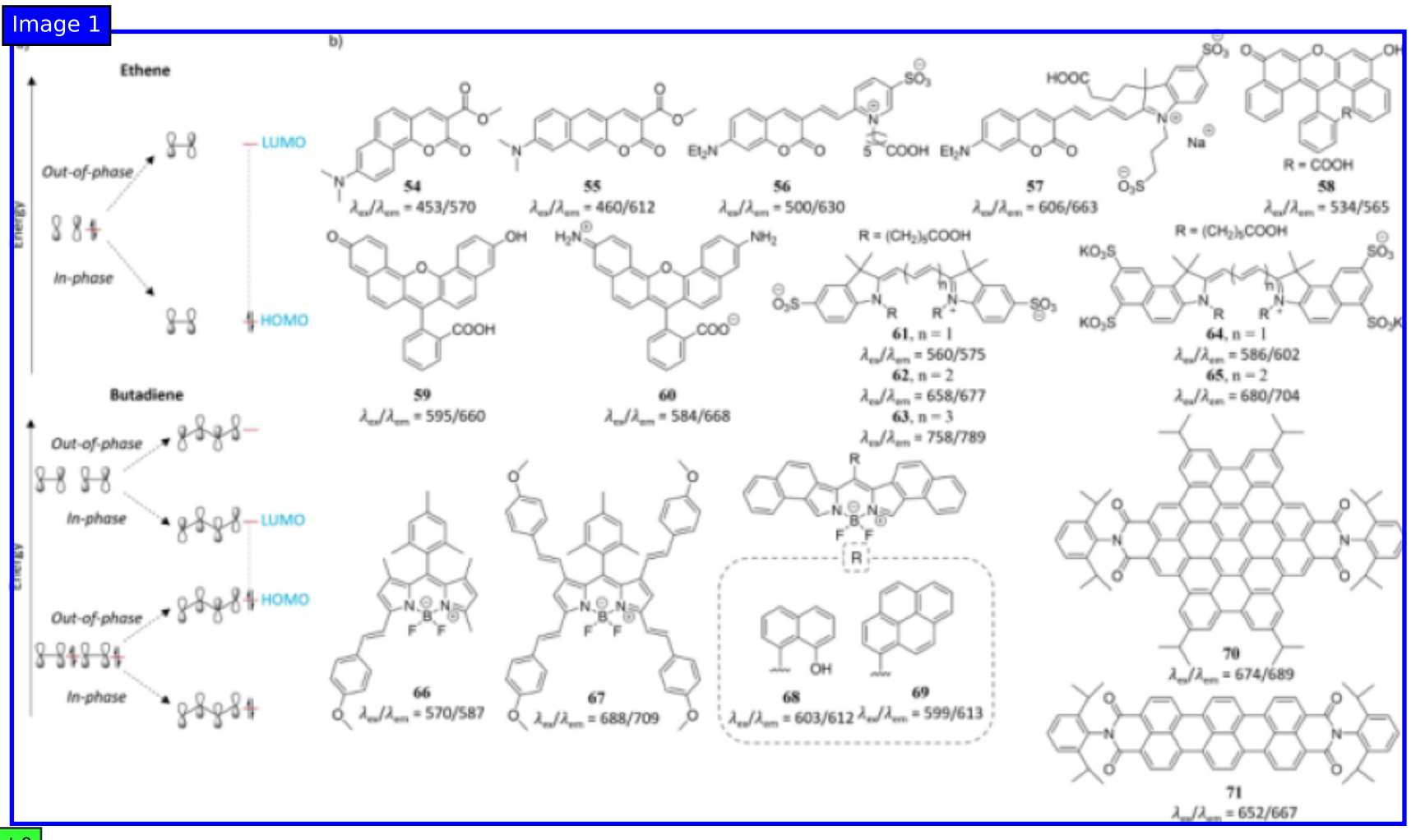
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one diimide that are highly tuneable, but are not covered here.[53]

Text 5
The *ortho*-position shows mainly LUMO contribution, therefore EWGs should red-shift the fluorescence. However, cyano substitution in those positions (**53**) cause a blue-shift.[54] TD-DFT calculations predict emission to be 5 nm red-shifted for **53** versus **48** and the fluorescence transition is LUMO→HOMO for both. Some have speculated a rational, but lack of experimental studies does not allow a conclusive verdict.[49] Red-shifting the PDI scaffold is achieved by putting strong EDGs in the bay-positions.

Text 6
2.2. Extending Conjugation

Text 7
Another common method to red-shift fluorophore emission is through π-conjugation extension, which stabilizes the LUMO and destabilizes the HOMO. This is explained by linear combination of atomic orbital LCAO theory, where molecular orbitals are formed by combining atomic orbitals of individual atoms in a molecule.[9,10] In π-conjugated systems, the π-electrons are delocalized over multiple atoms due to the overlapping of p-orbitals. These overlapping p-orbitals create a series of molecular orbitals consisting of bonding (lower energy) and anti-bonding (higher energy) orbitals.

Text 8
Consider ethene and butadiene LCAO, shown in Figure 8a. When a molecule has a shorter π-conjugated system, there are fewer p-orbitals involved in the overlapping, and the energy difference between the bonding and anti-bonding orbitals

Image 1


Text 9
Figure 8. a) Linear combination of atomic orbital depiction to illustrate why increasing conjugation length decreases the HOMO-LUMO gap, therefore, red-shifting emission. b) Examples of red-shifting fluorescence through conjugation extension of the π-system. **54**[55] in water; **55**[56] in pH 7.4; **56–57**,[57] **61–63**[58] in EtOH; **58–59**[40] pH 9.5; **60**[41] 1:9 dimethylsulfoxide:buffer pH 7.4; **64–65**[61] in methanol; **66–67**,[62] **68–69**[63] in dichloromethane; **70–71**[64] in chloroform.

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Text 1
(HOMO-LUMO gap) is relatively larger. On the other hand, in a molecule with an extended π-conjugated system, more p-orbitals are involved in the overlapping. As a result, the energy difference between the bonding and anti-bonding orbitals becomes smaller, causing a decrease in the HOMO-LUMO gap (butadiene, Figure 8a).

Text 2
Addition of benzofusions on coumarin shifts the fluorescence from 445 nm (**6**) to 570 nm (**54**) or 623 nm (**55**), depending on the fusion location.[65] Alternatively, aryl ethenylene moieties can shift the emission as well (**56**). There have also been coumarin-cyanine hybrids (**57**) synthesized that shift the fluorescence to 663 nm.[57]

Text 3
Benzofusions on fluorescein and rhodamine yield naptho-fluorescein and napthorhodamine, which shift the emission by 154 nm to 668 nm (**58–60**).[59,60] However, the pKa of naptho-fluorescein **60** is 8, therefore at physiological pH it is not in the highly fluorescent dianion form, limiting its usage.[6]

Text 4
Red-shifting cyanine emission through conjugation extension is well established. Either the methylene bridge is extended, or aryl fusions are added to the phenyl ring. Cy3 (**61**) has an emission of 575 nm. Extension of the bridge to Cy5 (**62**) shifts the fluorescence to 677 nm.[58] Further bridge extension to Cy7 (**63**) affords emission at 789 nm.[61] Unfortunately, bridge extension and red-shifting cyanine fluorescence comes at the cost of diminished quantum yields. Additional benzofusions to Cy3 and Cy5 shifts the fluorescence 27 nm to the red (**64** vs **61** and **65** vs **62**).[61]

Text 5
Aryl ethenylene addition on the BODIPY scaffold can dramatically shift the spectral properties. The more aryl ethenylenes incorporated, the further red the fluorescence.[62]

Text 6
For instance, one aryl ethenylene extends emission to 587 nm (**66**), while four of them push it even further to 709 nm (**67**).[62]

Text 7
In contrast, without this π-extension, the emission occurs at 519 nm (**34**).[36] Combining EDG and extending BODIPY conjugation can shift the emission to the near-infrared (780 nm).[66]

Text 8
Furthermore, conjugation can be extended off the pyrroles (**68** or **69**).[63] However, addition of aryl or ethenylene substituents at the *meso*-position does not result in significant red-shifting. The π-system in the *meso*-position is perpendicular to the π-system of the BODPIY core, therefore, is not in conjugation with the rest of the molecule (**62** versus **63**).[63]

Text 9
Red-shifting perylene emission is easily achieved through aromatic core extension. Perylene **48** emits at 536 nm, naphthalene addition to the core gives terrylene **71** emission at 667 nm.[64] Spectral shifting is also achieved via aryl fusions on the terrylene core (**71** vs **70**).[64]

Text 10
2.3. Heteroatom incorporation within the conjugated system

Text 11
Heteroatom incorporation into fluorescein, rhodamine, and BODIPY cores can also red-shift emissions where the more electronegative the atom, the better the electron withdrawing effect. BODIPY has significant LUMO density on the *meso*-position. Nitrogen incorporation in the *meso*-position of BODIPY **72** affords aza-BODIPY **73**. The fluorescence of **72** is shifted from 585 nm to 680 nm for **73**.[67] Here, the electronegative

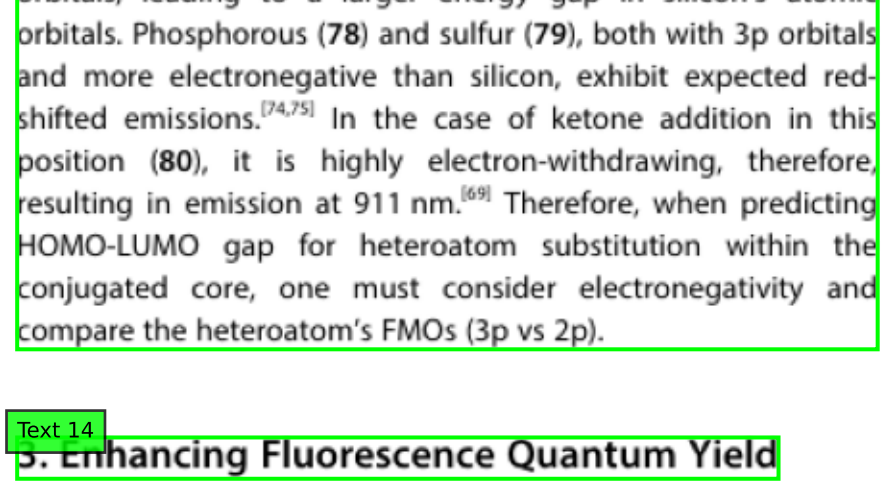
Text 12
atom lowers the LUMO significantly, which is responsible for the red-shift seen. Emission trends in rhodamine[68–77] and coumarins[78] follow a similar pattern, albeit with a few exceptions, highlighting the importance of other influencing factors.

Text 13
Replacing oxygen with a less electronegative nitrogen (**74**) leads to an expected blue-shift in the emission wavelength, from 562 nm to 528 nm.[70] However, substituting with a less electronegative carbon atom (76) surprisingly causes a red-shift in the emission. This is because nitrogen and oxygen have lone pairs that can donate into the pi-system, reducing their ability to stabilize the LUMO compared to carbon, despite being more electronegative. Interestingly, substituting carbon with silicon results in an even further red-shift for silicon-rhodamine **77**, with an emission at 648 nm, despite silicon's lower electronegativity.[71] This is because the energy levels of atom's p-orbitals (Figure 9b) must be considered, since the FMOs are derived from the linear combination of atomic orbitals (LCAOs).[79] Silicon possesses 3p orbitals, while carbon has 2p orbitals, leading to a larger energy gap in silicon's atomic orbitals. Phosphorous (**78**) and sulfur (**79**), both with 3p orbitals and more electronegative than silicon, exhibit expected red-shifted emissions.[74,75] In the case of ketone addition in this position (**80**), it is highly electron-withdrawing, therefore, resulting in emission at 911 nm.[69] Therefore, when predicting HOMO-LUMO gap for heteroatom substitution within the conjugated core, one must consider electronegativity and compare the heteroatom's FMOs (3p vs 2p).

Text 14
3. Enhancing Fluorescence Quantum Yield

Text 15
Quantum yield is a key factor in determining fluorophore brightness. Molecules that are not sufficiently bright are limited in their applications. To increase quantum yield, non-radiative decay pathways must be decreased.

Text 16
Fluorophores with significant conformational flexibility exhibit non-radiative energy loss after excitation and, therefore, low quantum yields. Scaffold rigidification reduces non-radiative decay and has been very successful for the cyanine scaffold.[80]

Image 1


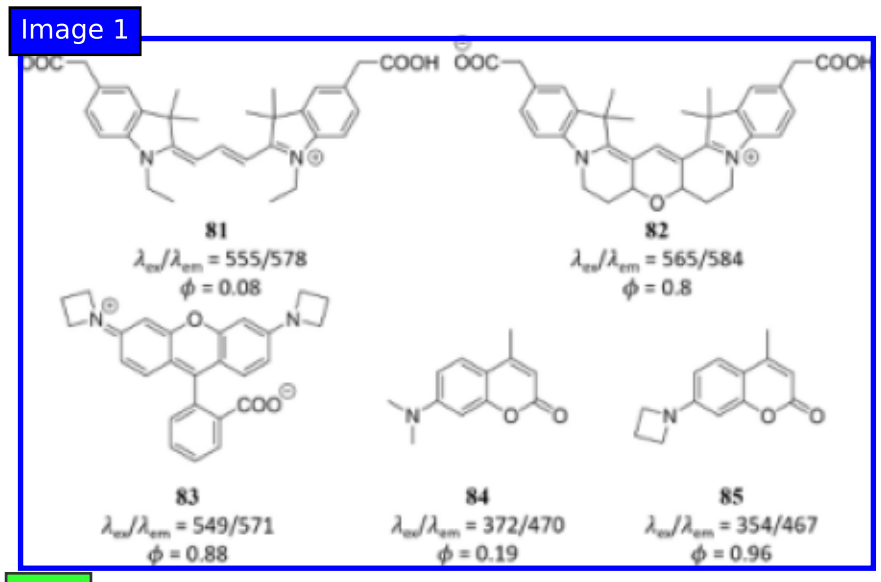
Text 17
Figure 9. a) Heteroatom incorporation to red-shift BODIPY fluorescence. b) Valence orbital energy level diagram to demonstrate they must be considered as they directly influence the energy gap. c) Red-shifting rhodamine fluorescence through modifying the heteroatom. **72–73**[67] are in 2-methyltetrahydrofuran. **74–80** full structures and solvents found here.[69]

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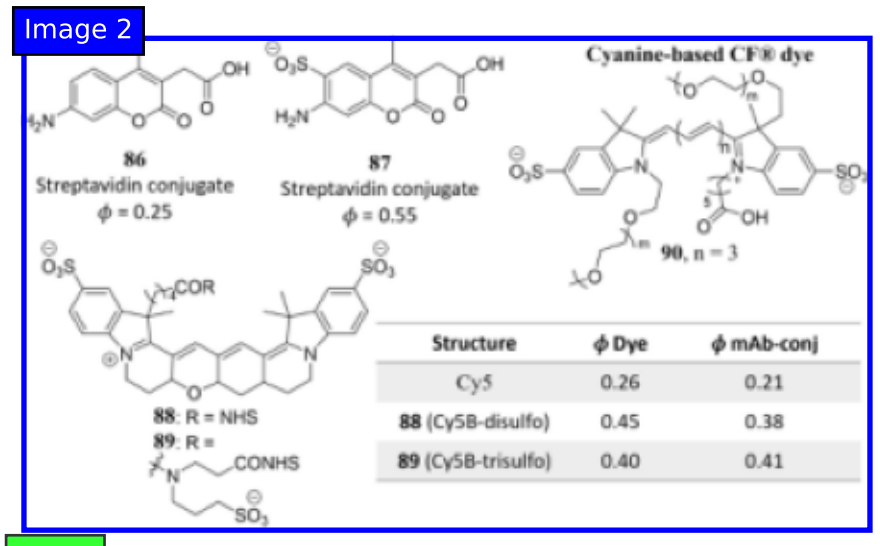
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Text 1
The quantum yield of cyanine dyes are decreased by photo-induced isomerization and intersystem crossing.[81] Rigidification of the polymethine bridge results in a minimization of non-radiative pathways and, therefore, an increase in quantum yield. Rigidification of cyanine **81** (Figure 10) results in a 10 fold enhancement in quantum yield for **82**.[82]

Text 2
Another non-radiative pathway that decreases quantum yields is twisted internal charge transfer (TICT), which shares similarities with ICT, but the main difference being TICT involves the twisting of the molecule's geometry during the charge transfer process. This is seen with Rhodamine 110 (**15**) and TMR (**16**). **15** has a quantum yield of 0.88, alkylation red-shifts the fluorescence, but is accompanied by a large decrease in quantum yield (0.41 for **16**). To mitigate TICT, Lavis and coworkers appended various sized nitrogen rings on rhodamine.[83] They found a 4-membered heterocycle (azetidinyl-rhodamine, **83**) more than doubled TMR **16** quantum yield, increasing from 0.41 to 0.88. These azetidinyl-rhodamines are known as Janelia Fluor® Dyes. The quantum yield is the same as the non-alkylated rhodamine **15**, but red-shifted by 50 nm. This approach was also effective for the amino coumarin scaffold. Dimethylamino coumarin **84** has a quantum yield of 0.19 and azetidinyl ring substitution affords **85** with a quantum yield of 0.96.

Image 1


Text 9
Figure 10. Quantum yield enhancement through structure rigidification and minimizing TICT. **81–82**[82] in methanol, **83–85**[83] pH 7.3.

Image 2


Text 10
Figure 11. Sulfonation to increase quantum yields. Streptavidin conjugates **86–87**[63] in buffer, **88–89**[85] pH 7.4.

Text 3
Even if the quantum yield is high in a cuvette, many times it is significantly quenched upon fluorophore conjugation to biologically relevant molecules, such as proteins. This quenching can arise from interactions between dye molecules, such as stacking or aggregation. A remedy for this quenching is sulfonation of the dyes. Sulfonation renders the dye negatively charged, decreasing the aggregation propensity. Sulfonation also increases solubility in aqueous media, decreasing aggregation induced quenching.[84] AMCA-streptavidin conjugate (**86**) has a quantum yield of 0.25 (Figure 11). Sulfonation of **2**, to give Alexa Fluor 350, and streptavidin conjugation gives **87** with a quantum yield of 0.55, more than double **86**.[84] This is the basis of the Alexa fluor dye series, where dye sulfonation improves performance in biologial applications. Schnerman and coworkers also observed a noteable difference when conjugating conformationally restricted cyanines to antibodies (**88** and **89**).[85] However, if the dye is too negatively charged due to sulfonate addition, there can be a significant increase in non-specific background when imaging. So balancing the addition of sulfonate groups with polyethylene glycol modifications can increase solubility and brightness, while decreasing non-specific background. This is the approach used for cyanine-based CF® dyes (**90**).[86]

Text 4
4. Improving Fluorophore Photostability

Text 5
High quality bioimaging requires bright and photostable fluorophores. The quality of cell images depends on the number of photons detected, also known as the "photon budget." A high photon budget is essential for obtaining detailed information. Photoinduced fluorophore degradation, or photobleaching, results in fluorescence termination, thereby reducing the photon budget. A major photobleaching pathway involves intersystem crossing to the triplet state followed by oxidation. Therefore, the two main methods used to enhance photostability are decreasing reactivity to reactive oxygen species (ROS) and shortening the triplet-state lifetime.

Text 6
Fluorination is a common approach to decrease reactivity to ROS because it increases the energy levels of the molecule, making it more difficult to oxidize.[45] Fluorescein fluorination decreases bleaching by half. For example, 17% of fluorescein fluorescence **11** is lost after 33 minutes of irradiation, but only 8% of fluorinated **91** fluorescence was lost.[28] Fluorescein photostability improves as the number of fluorines increases. This was also applied to coumarins and cyanines (Figure 12).[25,45] Additional modifications to improve photostability include azetdine incorporation in rhodamine as shown in **83**.[83]

Text 7
This is because the twisted ICT diradical intermediate could undergo irreversible bleaching reactions.[87]

Text 8
Alternatively, shortening the triplet state increases photostability. This is achievable with the addition of triplet-state quenchers (TSQs). Photophysical TSQs include cyclooctatetraene (COT) and nickel(II), which quench the triplet-state through energy transfer.[88,89] Reducing agents such as Trolox and β-mercaptoethanol (BME) act as photochemical quenchers *via* photoinduced electron transfer (PeT)-mediated

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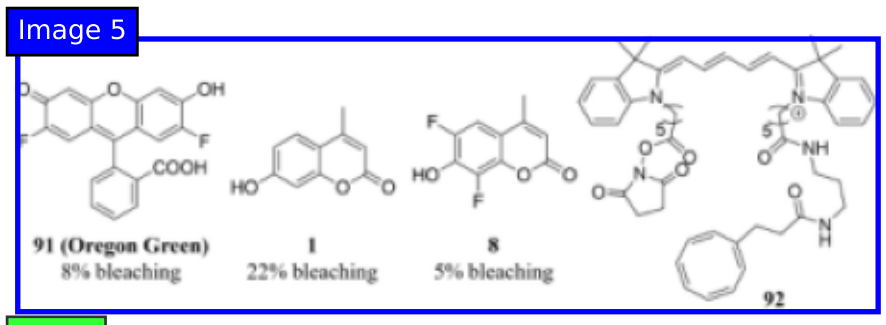
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Image 5



Text 14

Figure 12. Fluorination and attachment of TSQs to enhance photostability. **91**[28] at pH 9; **1**, **8**[25] pH 10; **92**[88] pH 7.5 under single molecule conditions.

Text 1

quenching.[88–91] Blanchard and coworkers were the first to covalently link TSQs to Cy5. The parent Cy5 showed 90% photobleaching compared to <20% of Cy5-COT **92**.[88] Significant advancements in long-term fluorescence imaging have been made by applying these methods.

Text 2

5. Newer Non-Planar Fluorophores

Text 3

Commonly used fluorophores are planar molecules discovered over 100 years ago. There has been much development of these structures, however, there is still a struggle with pH sensitivity,[92] brightness[6] and photostability.[93,94] As newer single-molecule and super-resolution imaging techniques become accessible, there is a greater demand for bright, photostable fluorophores. With advances in bottom-up synthesis, conjugated aromatic macrocycles are being realized as robust fluorescent sensors and labels. These unique structures offer properties unattainable with planar scaffold and expand the available chemical space.

Text 4

Aggregation-induced emission (AIE) fluorophores consist of conformationally flexible conjugated structures that remain non-fluorescent until they aggregate.[95] Conformational flexibility is constrained upon aggregation, promoting conjugation and, consequently, fluorescence. While AIE fluorophores can fall into the non-planar category, they have been a subject of extensive review for over two decades, including a dedicated ACIE special issue.[96–101] Therefore, this review does not cover AIE fluorophores. Instead, our focus is directed towards newer scaffold structures that have not been thoroughly explored in existing fluorophore reviews. This comparative analysis of emerging fluorophores against established counterparts plays a pivotal role in the advancement of innovative fluorescent molecules. We focus on squaraine-rotaxanes (sold by SETA biomedicals) and cycloparaphenylenes.

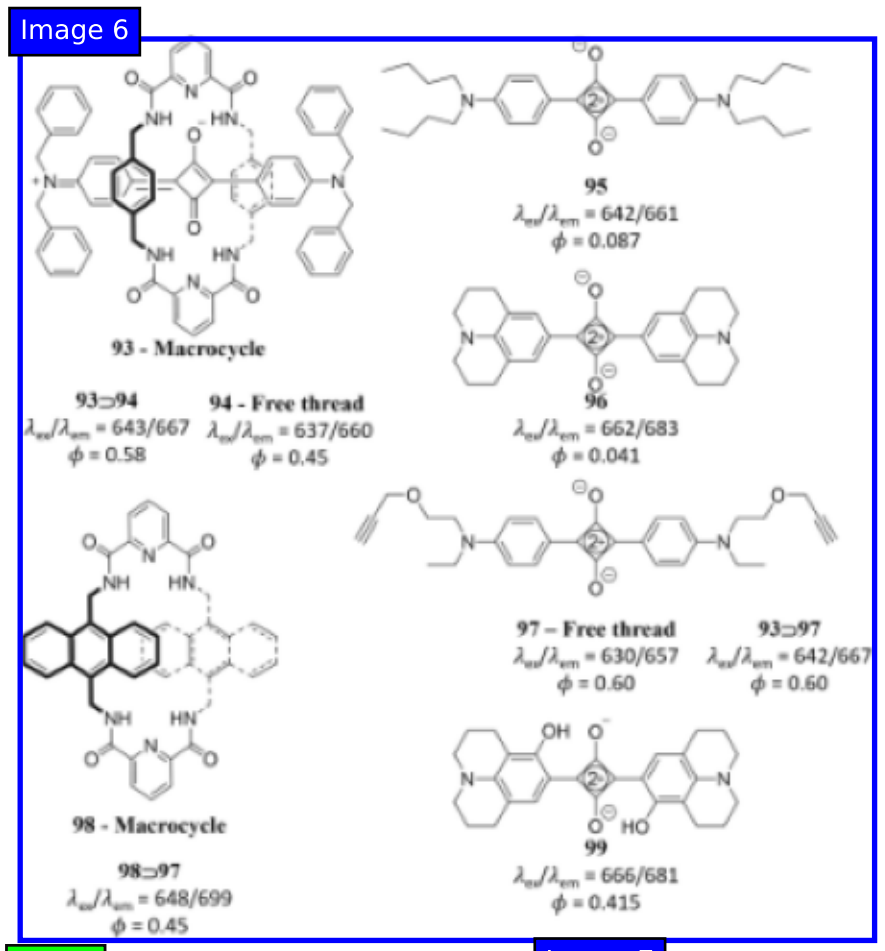
Text 5

5.1. Squaraine rotaxanes

Text 6

Squaraine rotaxanes are mechanically interlocked molecules comprised of a dumbbell (squaraine dye) inside a macrocycle (tetralactam), shown in Figure 13. Squaraines are donor-acceptor-donor ICT chromophores that absorb and emit in the far-red and near-infrared. Their emission properties make them highly desirable for biological applications. However, their utility is limited due to aggregation-induced fluorescence quenching in

Image 6



Text 16

Text 18

3. Squaraine[ ] 4:1 tetrahydrofuran:water;[102] **97**, **9**[ ] in chloroform;[103] **95–96**, **99**[104] in acetonitrile.

Text 7

aqueous environments and their susceptibility to nucleophilic attack. In 2005, Smith and co-workers addressed these issues by encapsulating a squaraine dye within a tetralactam macrocycle, creating the first squaraine rotaxane (**9**[ ]). These molecules are named with a [ ] between the macrocycle and thread respectively to signify that the thread is threaded through the macrocycle. While the squaraine fluorophore itself is not cyclic, the macrocycle necessary for its use as a biological fluorophore is. Moreover, the photophysical properties of squaraines are influenced by the macrocycle, warranting their categorization as non-planar fluorophores.

Text 10

The photophysical properties of squaraines arise from intramolecular charge transfer from the donor moieties to the C_4O_2 core. Therefore, tuning their photophysical properties involves modifying the donating and accepting abilities of these components. For instance, changing the alkyl amines in **88** to fused heterocycles like in **96** elicits a 22 nm red-shift in emission.[104–106] There are several reviews on methods to tune squaraine fluorescence.[102,104]

Text 11

An alternative method to alter the photophysical properties is by modifying the encapsulating macrocycle. The use of a tetralactam macrocycle leads to a 10 nm red-shift in the fluorescence of the free thread **97**, shifting emission from 657 nm to 667 nm (**9**[ ]). Further red-shifts in fluorescence are achieved through encapsulation within an anthracene macrocycle (**98**), resulting in a 32 nm shift for the **9**[ ] complex. The anthracene macrocycle reduces squaraine geometric deformations and stabilizes the excited state by electronic reorganization of the macrocycle.

Image 7
Image 9
Text 17
Image 1
Image 2
Image 3
Image 4
Text 8
Text 9
Text 12
Text 13

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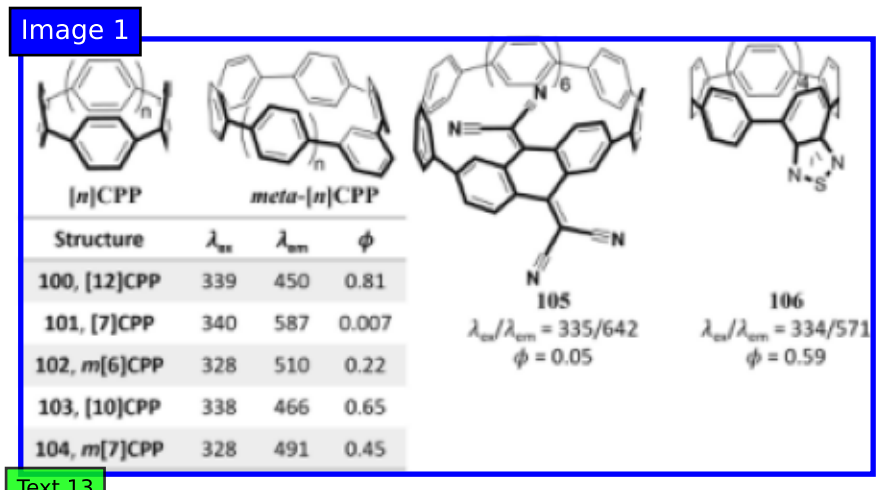
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Encapsulation of squaraines within macrocycles improves their quantum yields, increasing the quantum yield from 0.45 to 0.58 for **94**.[102,107–109] Additionally, introduction of substituents that restrict torsional motions around the squaraine single bonds, as seen with **99**, increases quantum yields. Phenols likely form hydrogen-bonds with the squaraine carbonyl groups, reducing the nonradiative decay, and increasing the quantum yield from 0.041 to 0.415.

Squaraine-rotaxanes are increasingly being utilized in biological imaging applications and are now available for purchase from SETA. These fluorophores have demonstrated low quenching effects when conjugated to biomolecules, along with exceptional photo- and chemical stability, particularly in single molecule microscopy.[110] Despite the progress made, there is still significant potential for exploring and fine-tuning the photophysical properties of squaraine-rotaxanes. As a result, we anticipate a growing surge in their usage and further advancements in this field.

5.2. Cycloparaphenylenes and Related Nanohoops

Cycloparaphenylenes (CPPs) are cyclic arrangements of *para*-linked phenylenes, resembling a slice of a carbon nanotube (Figure 14). Their cyclic structure provides enhanced solubility and distinct optical properties, setting them apart from linear fluorophores. All sizes of CPPs share a common absorption, making them ideal candidates as multiplexing dyes. Unlike planar fluorophores, CPPs exhibit a red-shift in fluorescence by reducing the number of phenylenes in the hoop. For example, [12]CPP (**103**) emits at 450 nm, whereas [7]CPP (**101**) emits at 587 nm. For these strained aromatic macrocycles, the important factors controlling the HOMO and LUMO levels are torsion angles between neighboring phenylenes and phenylene bending.[111] The smaller the CPP, the more bent the phenylene rings become, resulting in destabilization of the HOMO and stabilization of the LUMO. As CPP size is reduced, the torsion angle between neighboring benzene rings decreases. This results in an increase in π-conjugation, therefore, stabilizing the LUMO and destabilizing the HOMO. Together, bending and torsion effects result in a red-shift in fluorescence as the CPP size decreases.



Structure	λ_{ex}	λ_{em}	ϕ
100, [12]CPP	339	450	0.81
101, [7]CPP	340	587	0.007
102, *m*[6]CPP	328	510	0.22
103, [10]CPP	338	466	0.65
104, *m*[7]CPP	328	491	0.45

Figure 14. Cycloparaphenylene structures. **100**[113] in chloroform; **101**,[114] **102**,[114] **103**,[115] **104**,[116] **106**[115] in dichloromethane; **105**[117] in benzene.

However, when CPPs become too small, they are no longer fluorescent due to symmetry rules.[112] Disrupting the centrosymmetry by introducing a kink into the structure, *meta*-[*n*]CPPs, allows fluorescence in all sizes. [6]CPP is non-fluorescent, whereas *m*[6]CPP (**102**) fluoresces at 510 nm.

CPPs are essentially unaffected by electron donating and withdrawing group incorporation on their scaffold, since their HOMO and LUMO orbitals are evenly distributed. For CPPs, electron accepting moieties must instead be incorporated into the backbone.[118] Adding a tetracyanoanthraquinone unit (**105**) to [10]CPP (**103**) shifts fluorescence dramatically to 642 nm, though with a decreased quantum yield.[117] However, if the electron acceptor is switched to a benzothiadiazole (**106**) the fluorescence is shifted to 571 nm and the quantum yield is retained.[115] Furthermore, multiple benzothiadiazole units can further shift the fluorescence.[119] Numerous alternative and promising acceptors, as well as donors, have the potential to shift fluorescence into the near-infrared range; nevertheless, further synthesis is required to substantiate these possibilities.[118]

To enhance quantum yields in CPPs, different methods than those above are required. Deviating CPPs from their centrosymmetric nature allows LUMO to HOMO fluorescence transitions, resulting in higher quantum yields. Therefore, the larger and less strained the [*n*]CPP, the higher the quantum yield (**100** vs **101**). Additionally, disrupting the centrosymmetry, as in *meta*[*n*]CPPs, increases the fluorescence by 5—65 fold.[116] [7]CPP (**101**) is minimally fluorescent, with a quantum yield of 0.007. Changing one phenylene connection from *para*- to *meta*-connectivity yields *m*[7]CPP (**104**) with a quantum yield of 0.45.[116] Lastly, it is important to avoid ICT as it dramatically decreases nanohoop quantum yield.[115] While the quantum yield of **106** remains the same as the parent compound (**103**) because they fluoresce through the same mechanism, **105**, on the other hand, experiences ICT, resulting in a decrease in the quantum yield.

CPPs have shown potential as fluorophores for biological applications.[120–122] With recent advancements in functionalization, synthesis, and understanding of their photophysical properties, we expect that these distinctive structures to be further explored and adopted as fluorophores for fluorescence imaging. Their unique structure opens exciting opportunities, such as utilizing their cavities for supramolecular recognition and creating rotaxane dyes. CPPs hold promise as candidates for various imaging techniques in biological contexts.

6. Summary and Outlook

Fluorophores play a critical role in biology and biochemistry. Small molecule fluorophores are advantageous because precise manipulation of their structure alters their photophysical properties. We explored various methods for tuning fluorescence and examined fundamental physical organic chemistry concepts that govern these properties.

The key guidelines are:

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1. Consider frontier molecular orbital (FMO) distribution. Adding donor or acceptor groups to positions where only one FMO is localized, a specific FMO is electronically modulated without affecting the other. This changes the HOMO-LUMO gap and, therefore, fluorescence emission wavelength.
2. Electron-donating groups (EDGs) destabilize FMOs, while electron-withdrawing groups (EWGs) stabilize them. To red-shift fluorescence, add EDGs to positions with mainly HOMO orbital contributions and EWGs to positions with mainly LUMO contributions.
3. Increasing the conjugation within a molecule reduces the HOMO-LUMO gap due to linear combination of atomic orbital theory. When more p-orbitals participate in the overlapping of a conjugated system, the energy difference between the bonding and anti-bonding orbitals becomes smaller.
4. Heteroatoms can alter emission wavelengths. When predicting HOMO-LUMO gap for heteroatom substitution within the conjugated core, first use electronegativity to predict a trend, then adjust the trend by comparing valence orbital energy levels of the heteroatom.
5. Quantum yields are improved by structural rigidification, which minimizes non-radiative energy loss (ex. TICT), and increasing solubility through methods such as sulfonation to reduce intermolecular quenching.
6. Fluorination improves fluorophore photostability because it decreases fluorophore reactivity to reactive oxygen species (ROS). Additionally, shortening the triplet-state lifetime through triplet-state quencher attachment improves photostability. Lastly, fluorophore encapsulation and formation of a rotaxane can increase photostability.

With a better understanding of these physical organic chemistry concepts, we can move towards a more systematic fluorophore approach. Organic chemistry innovations will likely lead to new fluorophores with enhanced photophysical properties. Squaraine-rotaxanes and cycloparaphenylenes show promise as bright, stable alternatives to classic scaffolds, although they are still in early stages of development.

Creating fluorophores that combine all desirable properties remains challenging, as optimizing brightness and red-shifting fluorescence often leads to low quantum yield, insolubility and limited cellular permeability. Therefore, future fluorophore development should aim to incorporate these properties effectively. Improved fluorophores will undoubtedly advance our understanding of biological systems, uncovering new discoveries in the process.

Acknowledgements

I'd would like to acknowledge Dr. Curtis E. Colwell for his thoughtful discussions. This work benefited from access to the University of Oregon high performance computer, Talapas. RJ thanks the U.S. National Science Foundation for generous funding supporting nanohoop research (CHE-2102567, CHE-2204123).

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords: fluorophores · fluorescence · physical organic chemistry · fluorescent probes · photophysical properties

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Text 9

Text 10
Manuscript received: December 21, 2023
Revised manuscript received: January 8, 2024
Accepted manuscript online: January 11, 2024
Version of record online: February 2, 2024

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